

Skillomate Phase 7

Final PhonePe test-data cleanup and permanent production kill switch

CLEANUP	KILL SWITCH	DEPLOYMENT
PASS - production audit is zero	PASS - implemented and tested	NOT PERFORMED

Executive comparison

Area	Previous: Phase 7B	New: Final Phase 7
PhonePe data	Legacy users and payment records retained for compatibility.	Confirmed test population removed; final production audit reports zero PhonePe users and records.
Initiation	New checkout blocked by default unless an opt-in flag was enabled.	Requires both PHONEPE_ENABLED=true and the existing new-payment opt-in. Default is disabled.
Callback/webhook	Routes remained active for legacy obligations and used atomic multi-instance claims.	Routes return before database, provider, claim, payment, or subscription side effects when disabled.
Gateway routing	New checkout failed closed for missing or unknown mode.	All shared payment dispatch now fails closed for missing or unknown mode; Razorpay remains explicit.
Operations	No cleanup command or scoped backup workflow.	Backup-first, checksum-verified, exact-ID transactional cleanup with safe rerun behavior.

Outcome

The final audit reports 0 active PhonePe subscriptions, 0 pending PhonePe subscriptions, 0 pending PhonePe orders, 0 PhonePe test users, and 0 orphaned PhonePe identity references. No Razorpay, Google Play, Apple, course-content, or unrelated production records were selected by the cleanup plan.

Production data cleanup

The approved cleanup used exact MongoDB document identifiers captured in the verified backup. The deletion ran inside one transaction with snapshot reads and majority writes. User documents, when present, are ordered last.

Metric	Initial dry run	Final audit
Existing PhonePe test users	40	0
Candidate identity references	68	0
Orphaned identity references	28	0
Active PhonePe subscriptions	43	0
Pending PhonePe subscriptions	0	0
Pending PhonePe orders	26	0
Historical PhonePe orders	79	0
Historical PhonePe events	77	0

Backup and transaction evidence

Check	Result
Location	Outside repository under /private/tmp with directory mode 700
Archive	Compressed EJSON backup with file mode 600
Verification	SHA-256 checksum and 199-record re-read both passed before deletion
Transactional deletion	199 remaining records removed: 101 analytics, 4 certificates, 7 course-progress, 41 orders, 27 subscription events, 19 subscriptions
Rerun	Returned alreadyClean=true and deleted zero additional records

State-change note: the 40 User documents present during the approved dry run were already absent at the final pre-delete scope check, before this workflow invoked the cleanup command. The cleanup therefore backed up and transactionally removed only the remaining 199 confirmed records. The source of that intervening state change was not determined by this workflow.

Code and behavior changes

Component	Final Phase 7 change
Cleanup audit	Added a sanitized, read-only production audit that reports only counts and collection names and detects cross-provider conflicts.
Cleanup executor	Added explicit confirmation, expected-user count, out-of-repository backup, checksum verification, exact-ID deletion, transaction use, and post-cleanup audit.
Policy	Added PHONEPE_ENABLED. Missing or false disables PhonePe; new checkout also requires PAYMENT_GATEWAY_MODE=phonepe and PHONEPE_NEW_PAYMENTS_ENABLED=true.
Controllers	PhonePe callback and webhook return 410 before signature checks, claims, provider calls, model reads, or writes when disabled.
Routes	Shared cancel/status/access dispatch selects Razorpay explicitly and returns 503 for unknown or missing gateway modes instead of falling back to PhonePe.
Configuration	.env.example documents PHONEPE_ENABLED=false as the permanent production default.
Compatibility	Historical PhonePe schemas and multi-instance claim code remain available without being reachable for side effects while disabled.

Verification

Suite	Result	Interpretation
Focused Phase 7 and Razorpay	42 passed / 0 failed	Kill switch, no-side-effect behavior, fail-closed routing, backup, cleanup rerun, claims, and Razorpay regression coverage passed.
Complete backend	241 passed / 3 failed	Three known unrelated failures remain: two annual-billing harness expectations and one PDF worker dependency-resolution test.
Syntax and diff hygiene	PASS	Changed JavaScript parsed successfully and git diff whitespace checks passed.
Final production audit	PASS	All requested PhonePe cleanup counts and historical record counts are zero.

READY TO DEPLOY: YES

READY FOR MULTI-INSTANCE: YES

Deployment was intentionally not performed. Production must set PAYMENT_GATEWAY_MODE=razorpay and leave PHONEPE_ENABLED=false.